

ECOCIDE

A Satellite-Based Evidentiary Framework for War-Time Environmental Crimes

Executive Summary · DOI: 10.5281/zenodo.21757974 · Sakshi D. Maske

The Question

On 6 June 2023, Ukraine's Kakhovka Dam was destroyed, draining an 18.2 km³ reservoir and flooding downstream floodplain. Legal bodies are now considering “ecocide” as a prosecutable international crime, but existing satellite assessments of this event rely on visual, qualitative interpretation and explicitly decline to establish statistical causality. Can a causal-inference framework independently quantify conflict-attributable environmental damage, separated from Ukraine's already-elevated conflict baseline?

The Method

A Difference-in-Differences (DiD) model compares Kherson, Ukraine (treatment) against a matched non-conflict control zone — Tulcea, Romania / the Danube Delta (control) — with month fixed effects to isolate the conflict-attributable effect from seasonal cycles. Standard errors use the Newey-West HAC correction (not clustering, since this two-unit design has too few clusters for cluster-robust inference). Flood extent is sourced from UNOSAT's verified multi-sensor product (ICEYE, Landsat-9, SkySat, WorldView-3, MODIS) rather than derived independently from raw bands.

The Finding

A statistically significant, causally-validated NDVI decline was detected in Kherson relative to the control zone following the dam's destruction — confirmed through a clean placebo test using a counterfactual pre-event date.

Metric	Value
NDVI DiD Coefficient	-0.0703
P-value (HAC-robust)	0.022 — significant
95% Confidence Interval	[-0.130, -0.010]
Placebo Test #1 (fake date, June 2022)	+0.0148, p = 0.612 — clean pass
Peak Flood Extent (9 June 2023)	464.18 sq. km (UNOSAT, 5-sensor verified)

Validation & Robustness Checklist

- ✓ Matched non-conflict control zone — Tulcea, Romania / Danube Delta
- ✓ HAC-robust standard errors (Newey-West correction)
- ✓ Placebo Test #1 — clean pass on the broad baseline (primary result)
- ! Placebo Test #2 — genuine failure disclosed on a narrowed baseline (see limitation below)
- ✓ 23-quarter event-study validation
- ✓ Month fixed effects (seasonal controls)
- ✓ Multi-sensor verified flood data (UNOSAT, 5 independent sensors)

Honest Limitation

A quarterly event study found a significant effect in a pre-treatment quarter (summer 2022) — traced to Kherson already being an active conflict zone before the dam's destruction, meaning the original baseline wasn't a genuinely quiet pre-conflict period. A narrowed-baseline specification built to address this produces a larger effect (-0.1384, p = 0.0001) but its own placebo test fails once the correct HAC standard errors are applied (p = 0.001) — a genuine validation failure, disclosed as such and kept only as an illustrative sensitivity check. The broader-baseline result above, whose own placebo test remains clean, is the project's sole primary finding.

Real-World Relevance

International courts have already accepted satellite evidence in war-crimes prosecutions — the ICC's *Al Mahdi* case was built on satellite imagery of cultural-heritage destruction — and Ukraine's own Criminal Code (Article 441, enacted 2001) already criminalizes “mass destruction of flora and fauna” causing ecological disaster. This project applies the same causal-inference discipline used in policy evaluation to that evidentiary question, at the standard open-source investigators like Bellingcat and Human Rights Watch hold themselves to.

GitHub: github.com/sakshimaske303-commits/ECOCIDE | **Live Dashboard:** ecocide-xbub2cwcqjx9rkdd6nk5j5.streamlit.app |

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